Longest Common Subsequence

Topic: DynamicProgramming File: dp3.pdf

PROBLEM

Find the length of the longest sequence that appears in both strings in the same order; characters need not be adjacent.

STATE

dp[i][j] is the LCS length for prefixes of strings X of length i and Y of length j.

TRANSITION

If X[i-1] == Y[j-1], use 1 + dp[i-1][j-1]. Otherwise use max(dp[i-1][j], dp[i][j-1]).

EXAMPLE

X = ABCDGH and Y = AEDFHR have LCS ADH, with length 3.

COMPLEXITY

For string lengths m and n, time is O(mn) and the standard table uses O(mn) space.

APPLICATION

LCS is used in text comparison and forms a basis for related sequence-difference techniques.